

2.4.3 The for Loop

A common structure called a for loop is specially designed to manage counter-controlled looping.

```
for( int x = 1; x < 10; x++ )
```

An example program implementing the same:

```
// ForCounter.java
```

```
import java.awt.Graphics;
```

```
import javax.swing.JApplet;
```

```
public class ForCounter extends JApplet
```

```
{
```

```
public void paint( Graphics g )
```

```
{
```

```
for( int counter=1 ; counter <= 10 ; counter++ )
```

```
{
```

```
g.drawLine( 10, 10, 250, counter * 10 );
```

```
}
```

```
}
```

```
}
```

When appropriate, `for` is quick and easy. The `for` loop is a do-while. It tests the condition before it executes the loop for the first time.

Note: since the variable `int counter` was declared within `for`, it vanishes after the `for` is executed. All three sections are optional.

If you omit the condition, Java assumes the statement is true, and you have an infinite loop.

```
for( int x = 1;; x++ )
```

You can omit the initialization if you have initialized the control variable someplace else.

```
int x = 1;
```

```
for(; x < 10; x++ )
```

You can omit the increment of the variable if you are doing so within the body of the loop.

```
for( int x = 1; x < 10; )
```

```
{
```

```
other stuff
```

```
x++;
```

```
}
```

Let's take another example, this time a more comprehensive one to explain the use of the `for` loop.